

Bài 4

```
class HangSanXuat {  
    private String tenHangSanXuat;  
    private String tenQuocGia;  
    public HangSanXuat (String tenHangSanXuat, String tenQuocGia)  
    {  
        this.tenHangSanXuat = tenHangSanXuat;  
        this.tenQuocGia = tenQuocGia; }  
    public String getTenHangSanXuat () {  
        return tenHangSanXuat; }  
}
```

```
abstract class PhuongTienDiChuyen {  
    protected String loaiPhuongTien;  
    Protected HangSanXuat = hangSanXuat; }  
    public String layTenHangSanXuat ()  
    { return HangSanXuat.getTenHangSanXuat (); }  
    public void bat dau ()  
    { System.out.println (loaiPhuongTien + " khă hành "); }  
    public void tang toc ()  
    { System.out.println (loaiPhuongTien + " tang toc ... "); }  
    public void dung lai ()  
    { System.out.println (loaiPhuongTien + " đã dừng "); }
```



```

public abstract double layVantue ();
public class MayBay extends PhuongTiendichuyen {
    private String loaiNhienLieu;
    public MayBay (HangSanXuat, String loaiNhienLieu)
    {
        Super ("Máy bay", HangSanXuat);
        this.loaiNhienLieu = loaiNhienLieu;
    }
    public void catLanh ()
    {
        System.out.println ("cắt cánh");
    }
    public void hạCánh ()
    {
        System.out.println ("hạ cánh");
    }
}

```

@Override

```

public double layVanToc ()
{
    return 850.0;
}

```

```

public class XeOto extends PhuongTiendichuyen
{
    private String loaiNhienLieu;
    public XeOto (HangSanXuat, String loaiNhienLieu)
    {
        Super ("Xe Oto", HangSanXuat);
        this.loaiNhienLieu = loaiNhienLieu;
    }
}

```

@Override

```

public double layVanToc () { return 120.0; }
}

```



```

public class XeDap extends PhuongDienChuyen
{
    public XeDap (HangSanXuat)
    {
        super ("xe dap", hangSanXuat);
    }

```

@Override

```

    public double layVanToc () { return 15.0; }
}

```

```

public class Main {
    public static void main (String[] args)
    {
        HangSanXuat h1 = new HangSanXuat ("Bos", "USA");
        HangSanXuat h2 = new HangSanXuat ("Toyo", "Japan");
        MayBay mb = new MayBay (h1, "xăng");
        xe Oto oto = new xeOto (h2, "xăng");
        mb.bat dau ();
        System.out.println (+ mb.layVanToc () + "Km/h");
        mb.cat lanh ();
        oto.bat dau ();
        System.out.println (+ oto.layTenHangSanXuat ());
        System.out.println (+ oto.layVanToc () + "Km/h");
    }
}

```


Bài 5

```
public interface Moveable
{
    void moveUp();
    void moveDown();
    void moveLeft();
    void moveRight();
}

public class MoveablePoint implements Moveable
{
    int x, y, xSpeed, ySpeed;
    public MoveablePoint (int x, int y, int xSpeed,
                           int ySpeed)
    {
        this.x = x;
        this.y = y;
        this.xSpeed = xSpeed;
        this.ySpeed = ySpeed;
    }
}
```

@Override

```
public String toString()
{
    return "Point at (" + x + ", " + y + ") with  
Speed = (" + xSpeed + ", " + ySpeed + ")";
}
```

@Override

```
public void moveUp() { y -= ySpeed; }
```

@Override

```
public void moveDown() { y += ySpeed; }
```


@Override

```
public void moveLeft() { x -= xSpeed; }
```

@Override

```
public void moveRight() { x += xSpeed; }
```

```
public class MoveableCircle implements Moveable  
{ private int radius;
```

```
private Moveable Point center;
```

```
public Moveable Circle (int x, int y, int  
xSpeed, int ySpeed)
```

```
{ this.center = new moveablePoint(x, y, xSpeed,  
ySpeed);
```

```
this.radius = radius; }
```

@Override

```
public String toString() {
```

```
return "Circle " + radius + " center: " + center.  
toString(); }
```

@Override

```
public void moveUp() { center.moveUp(); }
```

@Override

```
public void moveDown() { center.moveDown(); }
```

@Override

```
public void moveLeft() { center.moveLeft(); }
```


@Override

```
public void moveRight () { center.moveRight (); }  
}
```

```
class Main {
```

```
public static void main (String [] args)
```

```
{ moveablePoint point = new MoveablePoint (10,10,5,5);
```

```
System.out.println (+ Point.toString ());
```

```
point.moveUp ();
```

```
point.moveRight ();
```

```
System.out.println (+ point.toString ());
```

```
moveableCircle c1 = new moveableCircle (20,20,2,2,50);
```

```
System.out.println (+ c1.toString ());
```

```
c1.moveDown ();
```

```
c1.moveLeft ();
```

```
System.out.println (+ c1.toString ());
```

```
moveable m1 = new moveablePoint (0,0,1,1);
```

```
m1.moveRight ();
```

```
System.out.println (" M1 (Point) : "
```

```
+ m1.toString ());
```

```
}
```

```
}
```